

Heat and Control, Inc.
Packaging Service Department
Technical Publication PS008

Atlas 202

Heater Sensor Assembly Modification

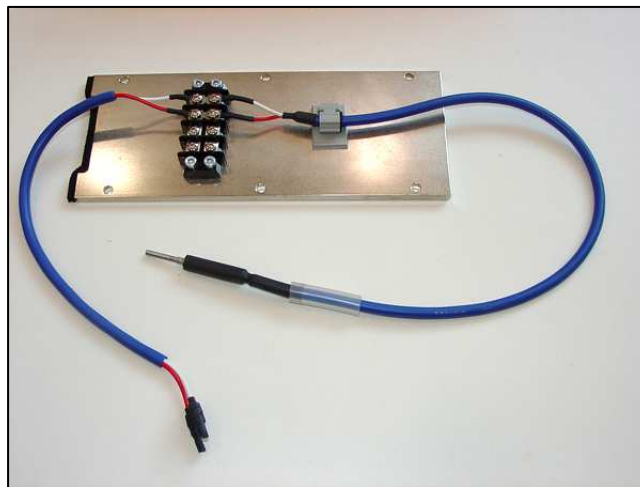


Photo 1: Atlas 202-Heater Modification Kit

Heater Assembly/Atlas-201Mod Kit for ATLAS-202

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Technical Publication PS008 Atlas 202 Heater Sensor Assembly Modification

WARNING! Follow Lockout /Tagout Procedure and allow the jaw area to completely cool prior to making this modification to the heater.

1. Tools Required:

Protective gloves
Allen wrench set: 1.5, 2, 3, 4, 5, 6mm
Phillips screwdriver
Adjustable wrench
Needle-nose pliers
Channellocks
Measuring tape



Photo 2: Required tools

2. Lockout / Tagout

Prevent accidental startup and injuries by securing the power cord in the lockout equipment.

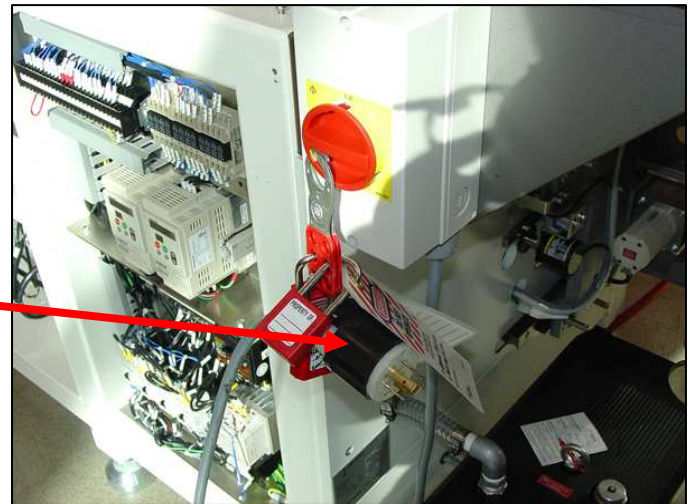
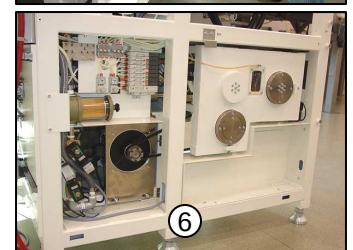
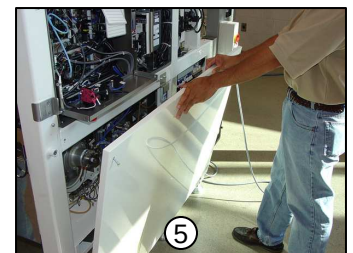
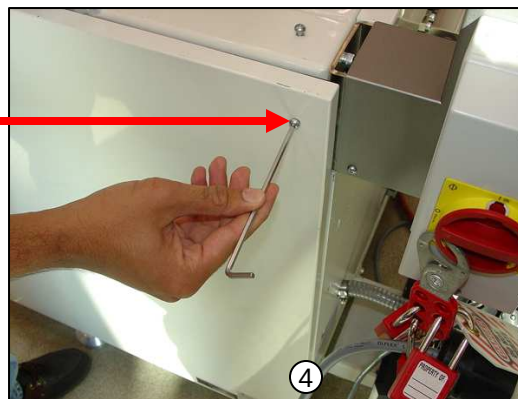


Photo 3: Lockout / Tagout equipment

3. Remove the Side Panels

Using a 4mm Allen wrench, remove the two panel bolts on both sides of the Atlas, lift the panels free and set them aside in a safe location where they will not fall.



Photos 4, 5, and 6: Side Panels removed



Photo 7: Jaw Motor Disconnect

4. Disconnect the Jaw Motor

Using your hand, disconnect the plastic elbow to the jaw motor which will permit the jaws to rotate smoothly and more safely. (the motor is found in the lower section in the rear of the left side compartment on a right-hand machine.)



Photo 8: Commutator Assembly

5. Locate the Commutator Assembly

This is the location where the new heater cables will be connected. It is found in the right side lower compartment on a right-hand machine and must be disassembled completely from both the front and rear jaws during this modification.

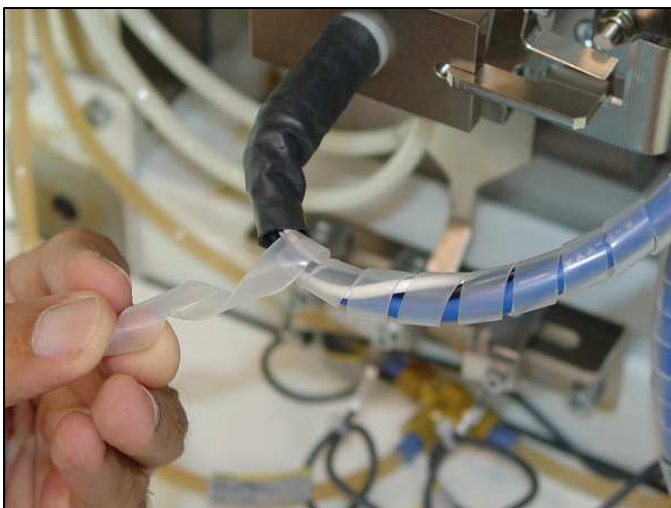
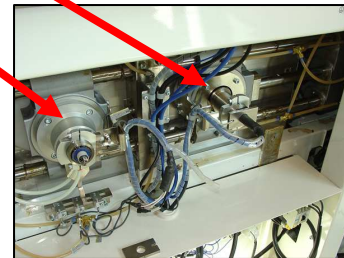


Photo 9: Unwinding the Spiral Wrap

6. Remove the Spiral Wrap

The protective spiral wrap that shields the commutator cable must be unwound for approximately 6".

7. Slide the cable connection boot back:

There are four wires within the cable that are color coded and must be attached to the correct pins. When the boot is slid back, the wires can be seen.



Photo 10: Cable connection boot

8. Note the position of the wires

(The wires will be removed during the modification and must be reattached exactly as they were connected originally.)

Disconnect the wires at this time.

Only remove the sensor cables.
did not remove power cables

Skip step #9

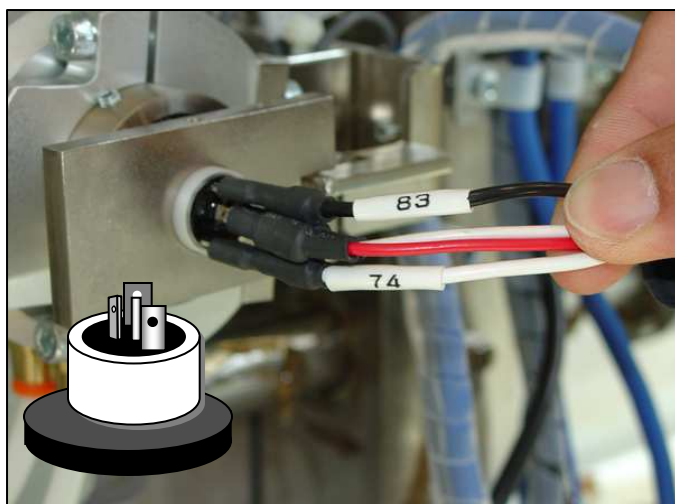


Photo 11: Sensor cable wires

9. Remove the End Plate

Using a 1.5mm Allen wrench, Loosen the two set screws that hold the rectangular plate on the end of the commutator. Remove the plate and set it aside.

Note: This modification will require a large number of small components to be taken on and off. Use caution to organize the temporary placement of these bolts, plates, washers, etc. to make reassembly easier.

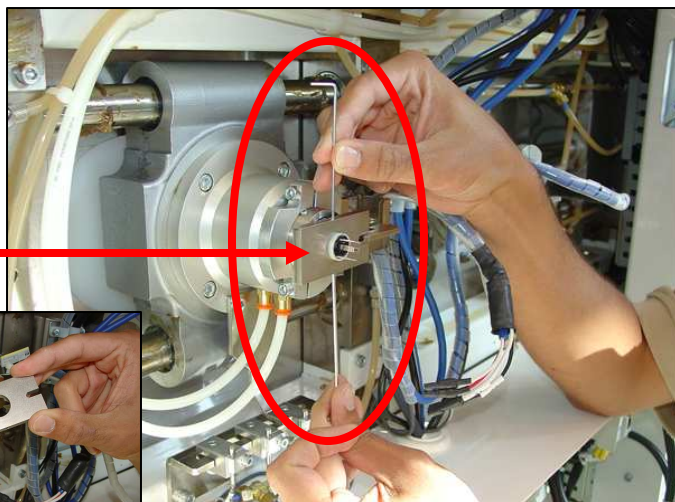


Photo 12: Loosening the set screws to remove the plate

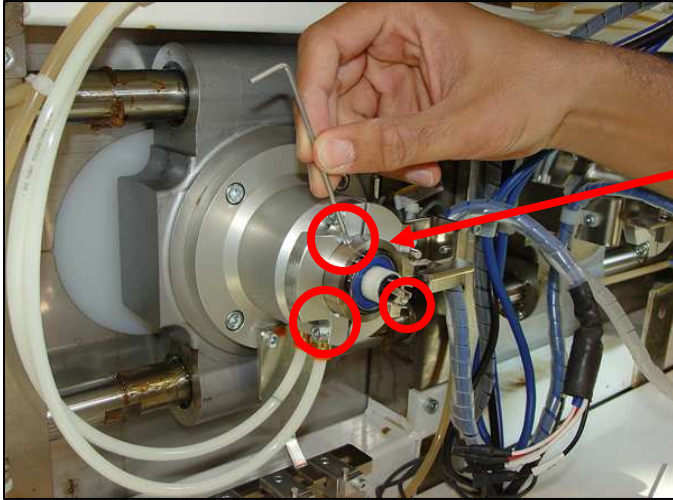


Photo 13: Commutator Set Screws

10. Loosen 3 Set Screws

Using a 2mm Allen wrench, loosen the three set screws which lock the commutator in place.

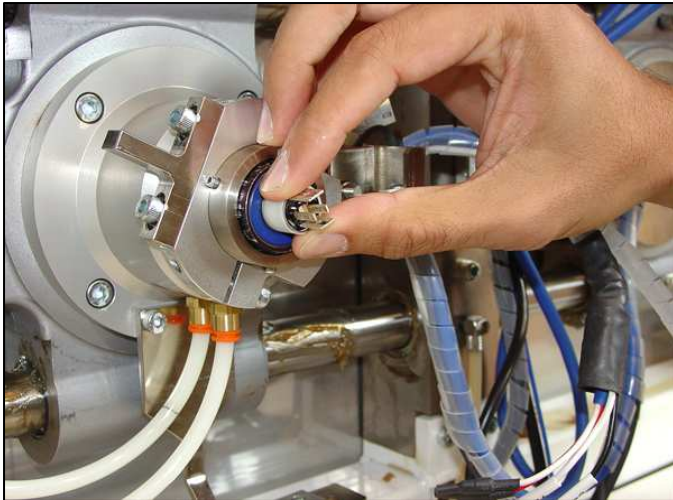


Photo 14: Checking for looseness

11. Check the Commutator is loose.

Do not attempt to remove the commutator yet.

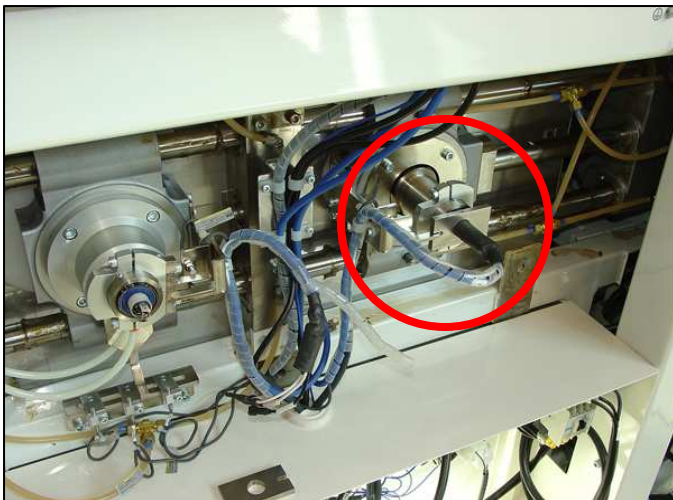


Photo 15: Rear Jaw location

12. Repeat Steps 5 through 11 for the Rear Commutator.

13. Open the Front Door to Access Jaws .

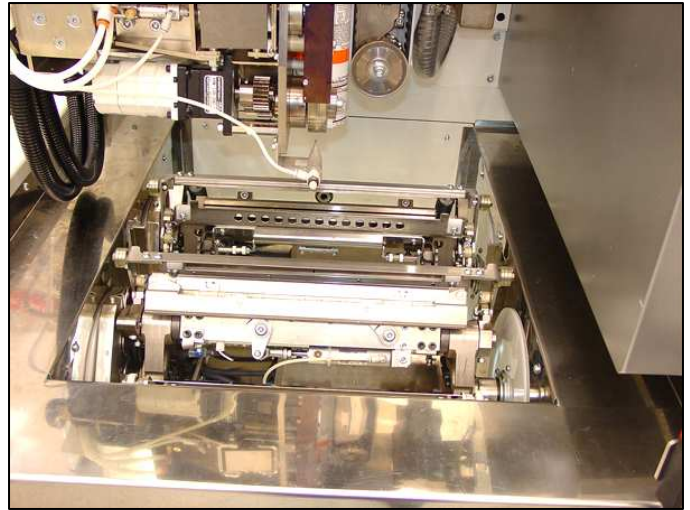


Photo 16: Front view of Jaws

14. Remove the Crumb Plate Bolts.

Using the 5mm Allen wrench
remove the 1st set of front and rear
crumb plate bolts.

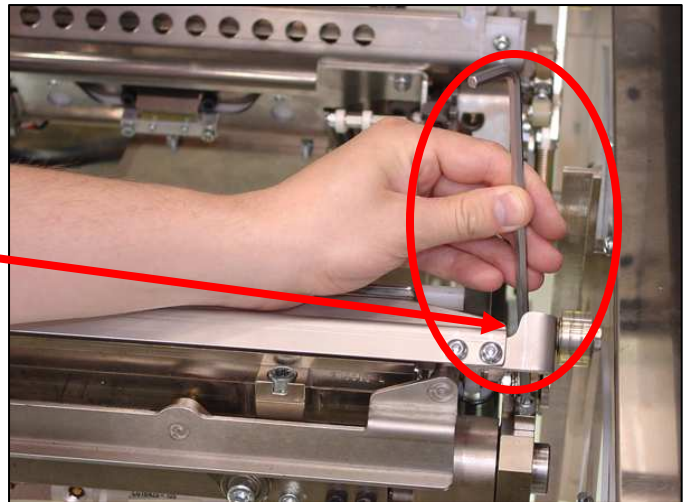


Photo 17: Removing the Crumb plate bolts

15. Remove the Crumb Plates and set aside.

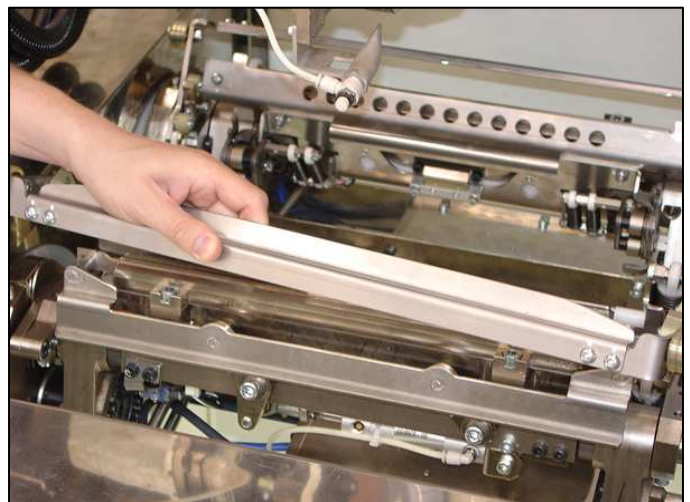


Photo 18: Removing the crumb plates

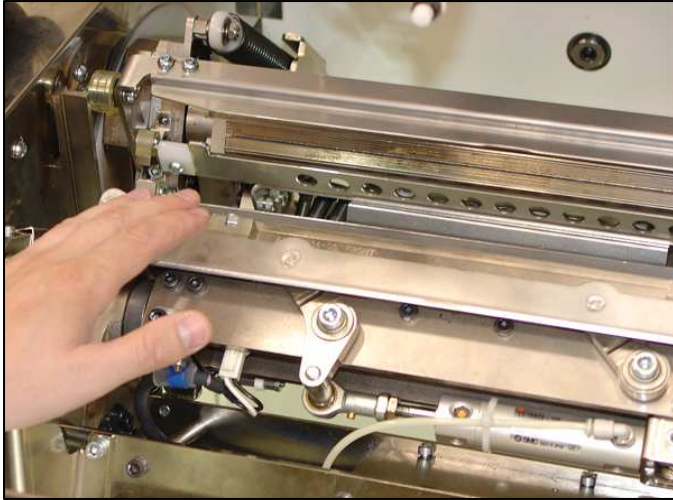


Photo 19: Rotating the jaws

16. Rotate the Jaws.

Using the 5mm Allen wrench to remove the 2nd set of front and rear crumb plates



Photo 20: Removing the stripper plate bolts

17. Remove the Stripper Plates.

Using the 5mm Allen wrench to remove the 1st set of front and rear stripper plates.

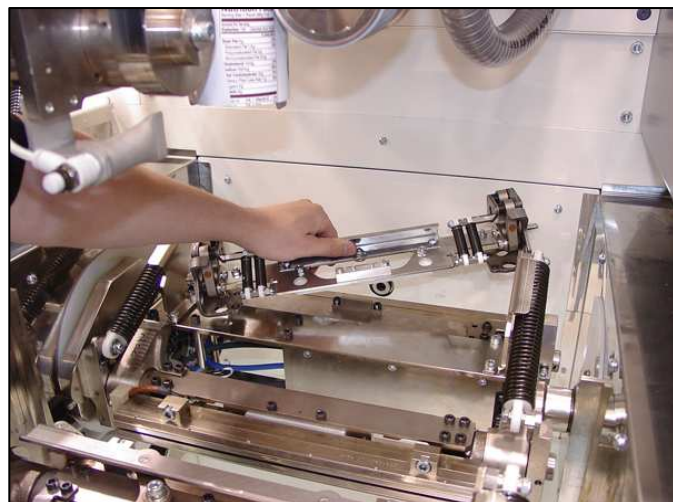


Photo 21: Removing the stripper plates

18. Set the Stripper Plates Aside.

19. Rotate the Jaws :

Using the 5mm Allen wrench to remove the 2nd set of front and rear stripper plates

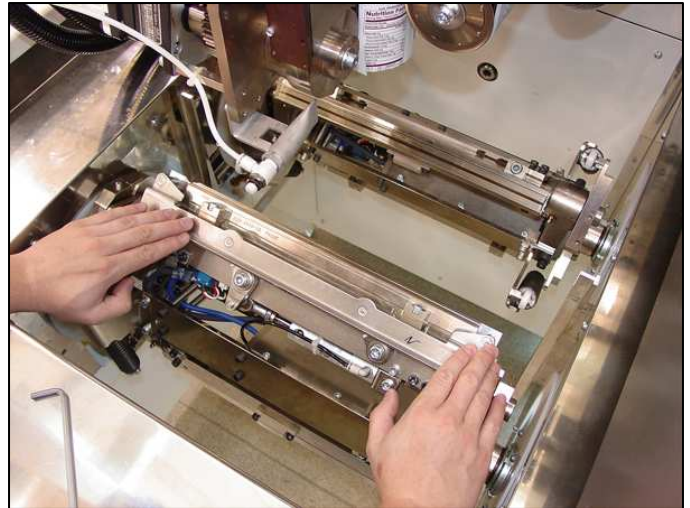


Photo 22: Rotating the jaws

20. Removing the Knife bolts

Using a 4mm wrench remove the 1st knife bolts.

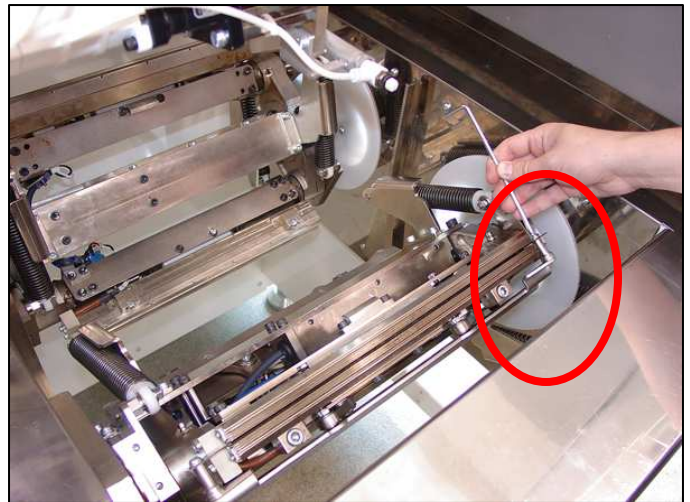


Photo 23: Removing the knife bolts

21. Carefully Remove the 1st Knife

Set the knife aside.

22. Rotate the Jaws to remove the 2nd Knife :

Carefully remove the knife and set it aside.

Caution: The Knife is very sharp.



Photo 24: Removing the knife



Photo 25: Unplugging the air hose

23. Unplug the Air-hose for the Knife

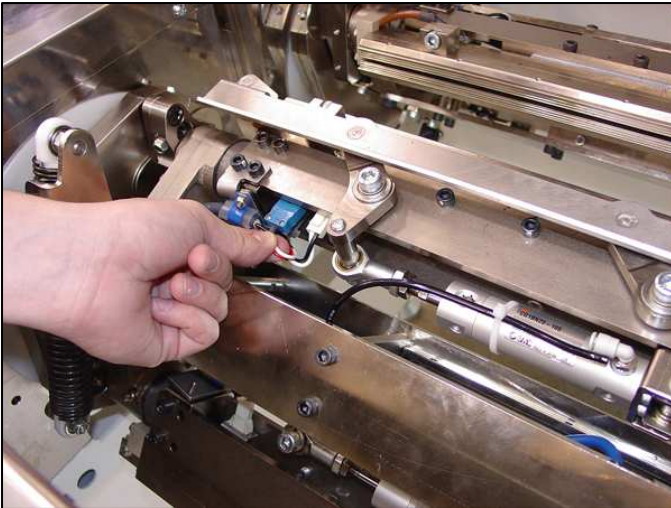


Photo 26: Unplugging the heater sensor

24. Unplug the Old Heater sensor

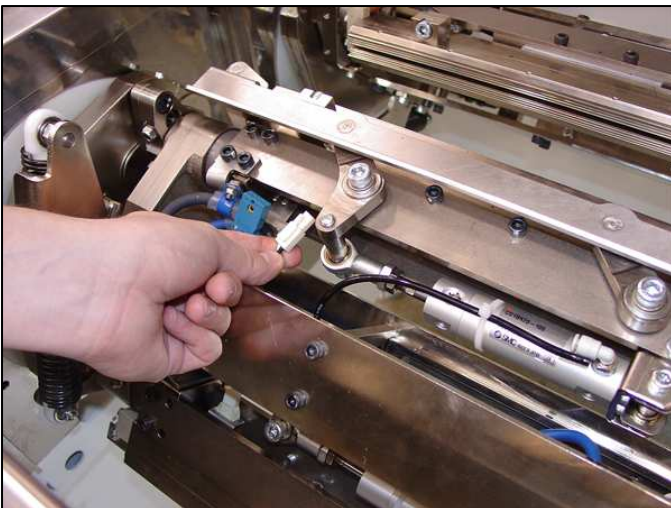


Photo 27: Unplugging the heater power

25. Unplug the Heater Power Connector

Note: remove the power connector from the front and rear jaws.

26. Rotate the Jaws:

Repeat steps 23, 24, and 25.

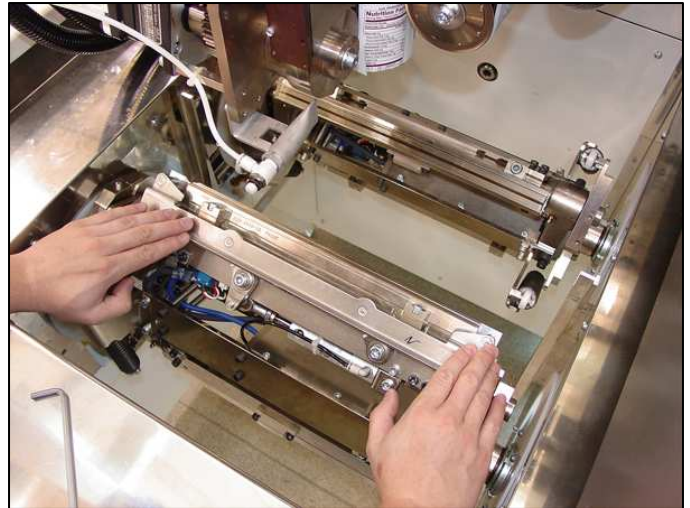


Photo 28: Rotating the jaws

27. Release the Jaw Tension Springs

Using a 5mm wrench release the front and rear tension spring bolts.

Caution: The springs are under high tension
Hold the springs securely.



Photo 29: Removing the tension spring bolts

28. Rotate the Jaws and Release the other J aw Tension Spring.

Photo 30: Releasing the tension springs

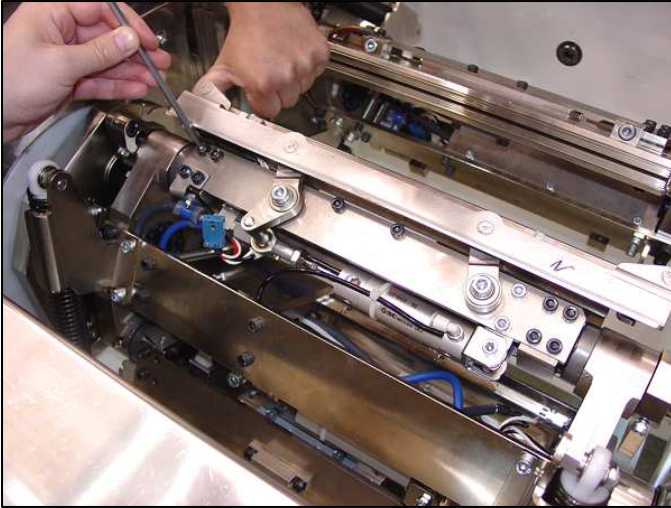


Photo 31: Removing the Jaw Bolts

29. Remove all four Jaw Assemblies

Using a 5mm wrench, remove the top jaw assembly bolts.

Only remove the JAWS with the sensor on the front and the rear

other 2 JAWS were left on machine

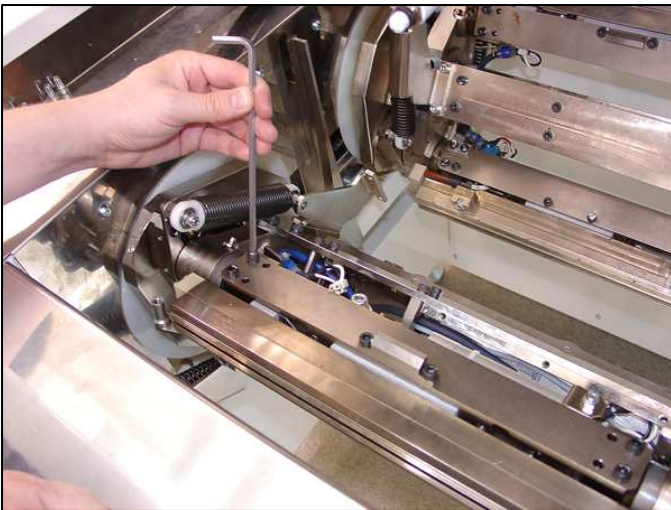


Photo 32: Removing the jaw bolts

30. Rotate the Jaws to access the bottom bolts.

Caution: When removing the remaining bolts, hold the jaws securely to prevent the jaws from dropping.

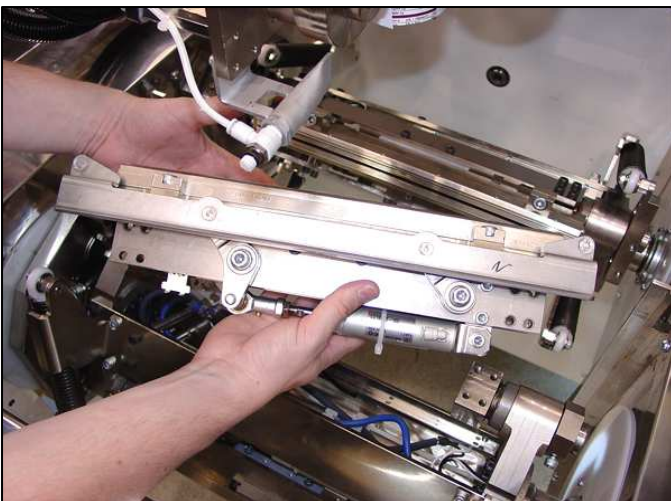


Photo 33: Removing the jaws

31. Remove the Jaws to a clean worktable for disassembly.

32. Disconnect the Sensor Cable from the Bracket:

Use a 7mm wrench to disconnect the cable.



Photo 34: Removing the Sensor Cable

33. Remove the Sensor Cable

Be careful not to drop the small washer, nut, and clip. They must be reused during re-installation.

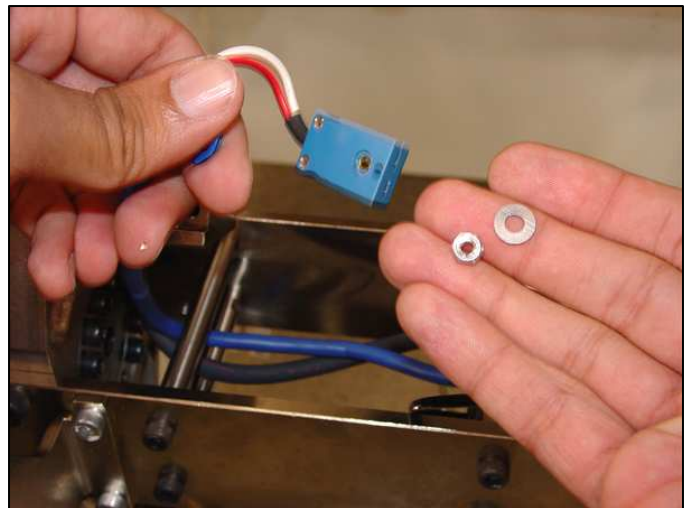


Photo 35: Removing sensor hardware

34. Remove the Cable Cover Plate Bolts

Using a 4mm Allen wrench, remove the six bolts.

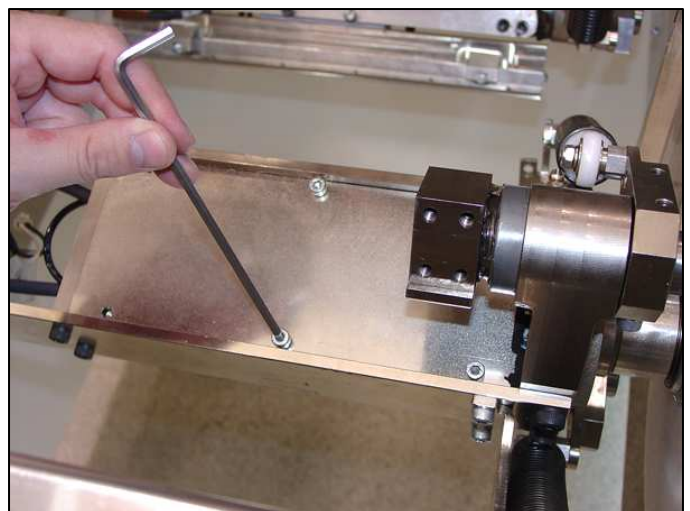


Photo 36: Removing the cables cover plate bolts

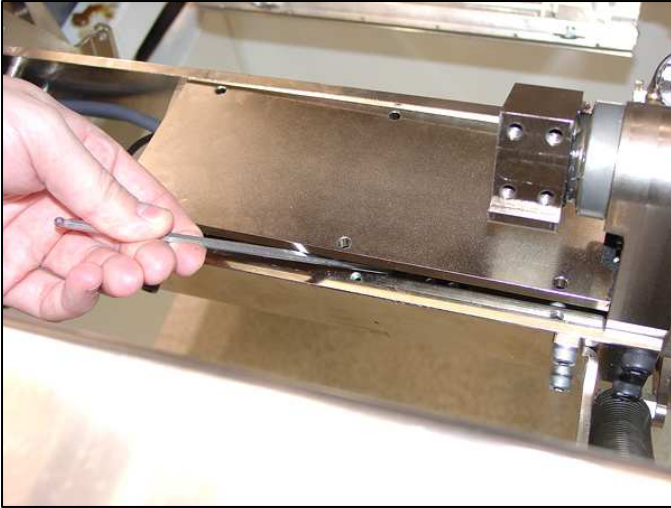


Photo 37: Removing the cables cover plate

35. Remove the Cables Cover Plate

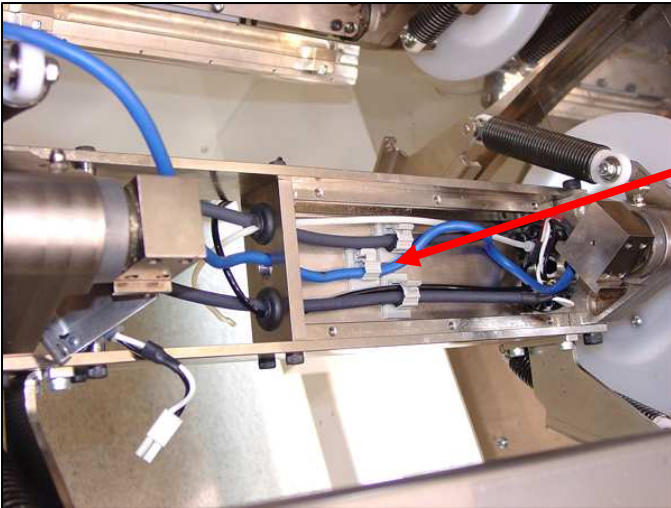


Photo 38: Releasing the cables

36. Release the three cables from their clips

This will allow the commutator to slide out.

only remove the Sensor cable
did not the power cables

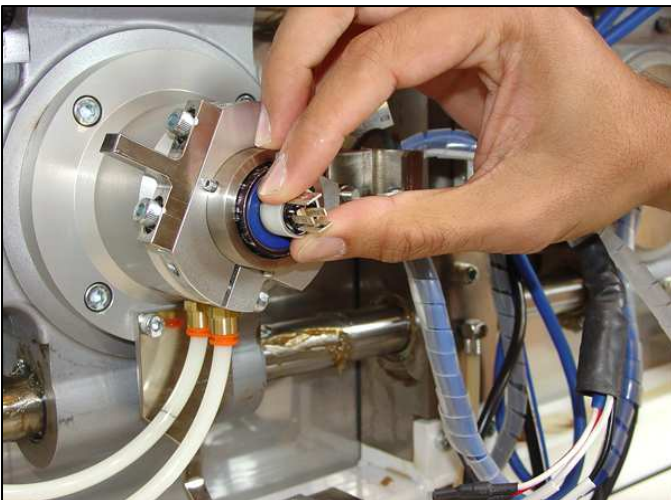


Photo 39: Removing the commutator

37. Remove the Commutator

This step may require the careful use of channellocks to free the commutator from its housing.

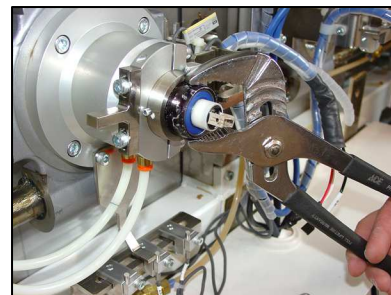


Photo 40: Removing the commutator

38. Note the position of the wires

(The wires will be removed during the modification and must be reattached exactly as they were connected originally.)

Disconnect the wires at this time.

only unplugged the sensor cables.
did not remove the power cables.

did step 61 at this time.
installed new plate and new sensor
cable to commutator



Photo 41: Unplugging wires from commutator

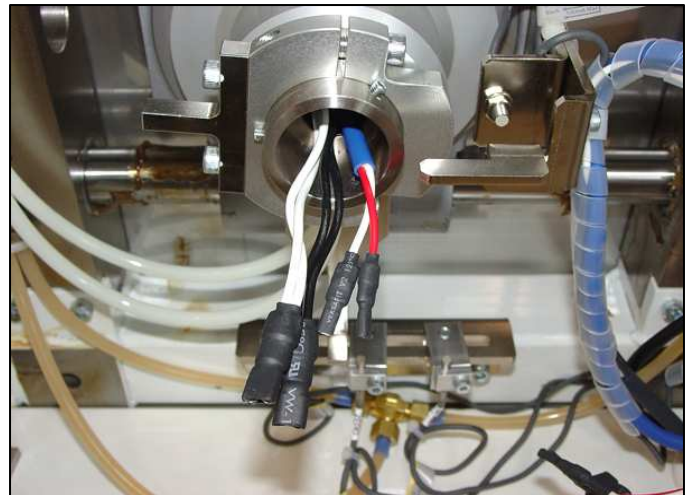
39. Remove the Commutator and set aside.

Photo 42: Removing the commutator

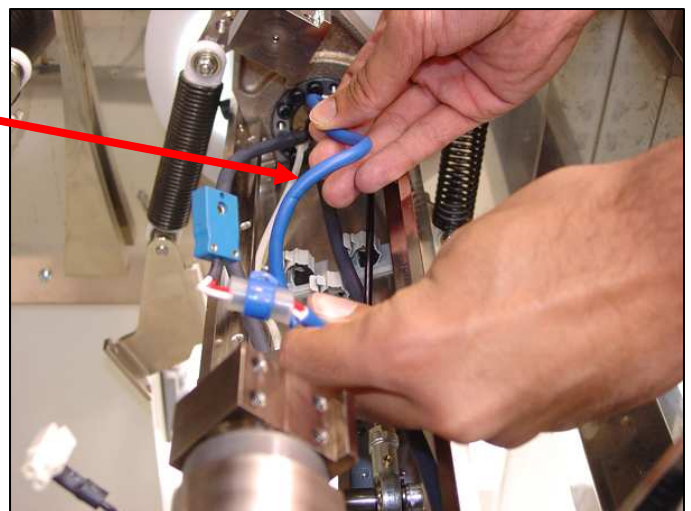
40. Remove the Blue Sensor Cable only and place it on the worktable.

Photo 43: Removing the old sensor cable

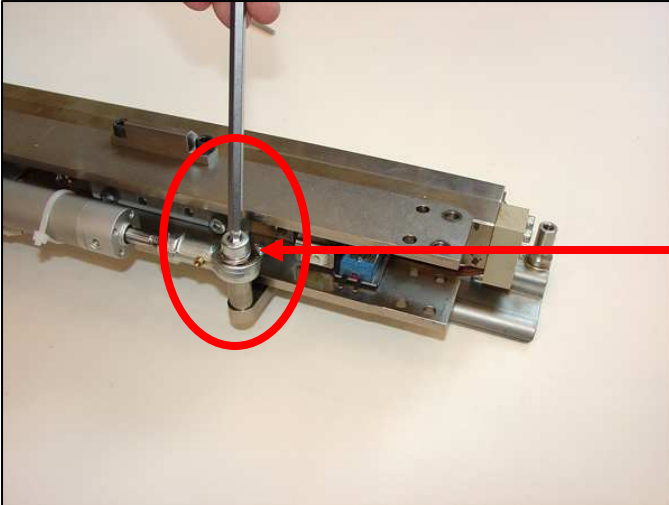


Photo 44: Removing the knife cylinder bolt

41. Remove the Knife Cylinder bolt from the Bracket

Using a 6mm Allen wrench, remove the cylinder bolt near the original sensor.

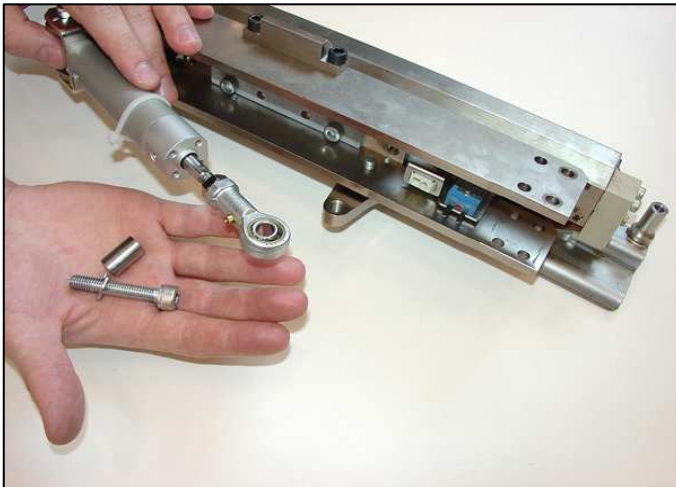


Photo 45: Disconnecting the knife cylinder

42. Disconnect the Knife Cylinder

Set the hardware aside for later reassembly

only remove one bolt , and swing bracket around.

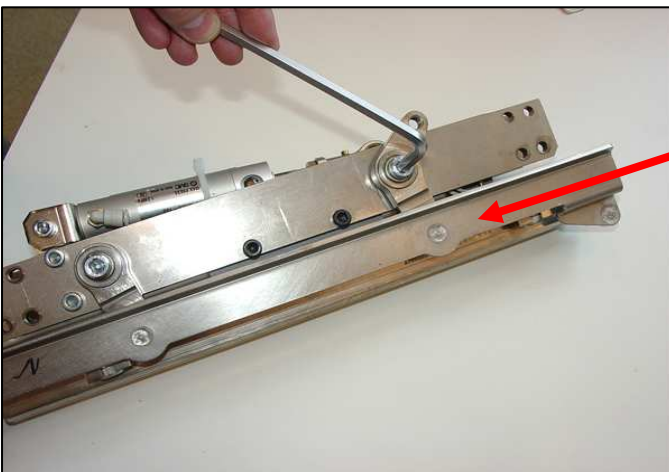


Photo 46: Removing the knife bracket bolts

43. Remove the Bracket from Jaw Assembly.

Using a 6mm Allen wrench, remove the bracket and set it aside.

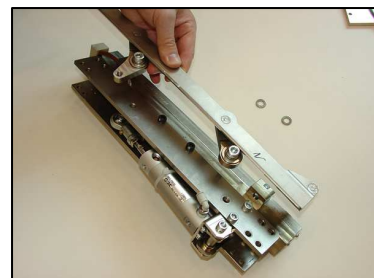


Photo 47: Removing the knife bracket

44. Remove the Sensor Clip

Use a 3mm Allen wrench to remove the bolts which will be used later

Discard the clip.

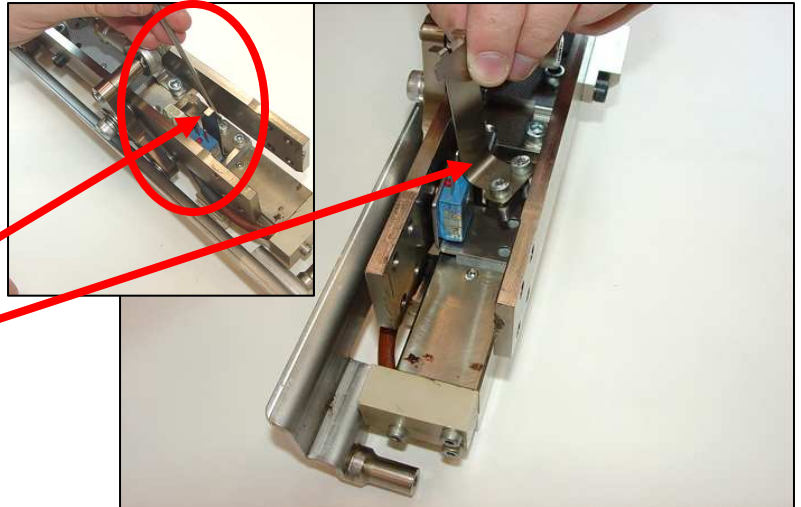


Photo 48: Removing the old sensor clip

45. Rotate jaw and remove set screw

Using 2mm wrench, remove the set screw.

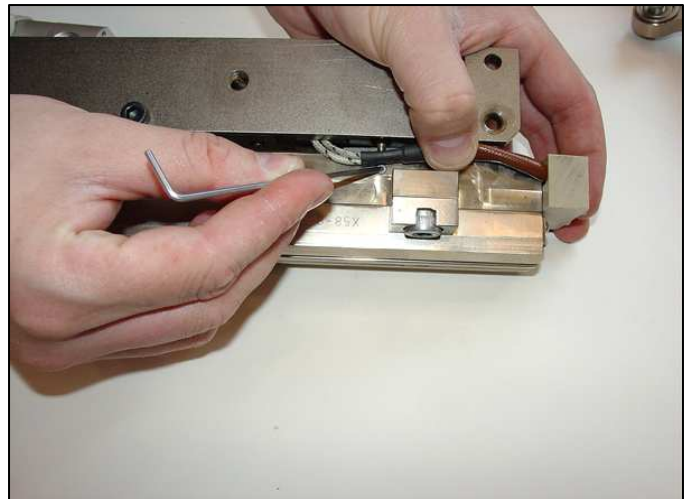


Photo 49: Removing the sensor set screw

46. Compress the sides of the White Connector using needle-nose pliers

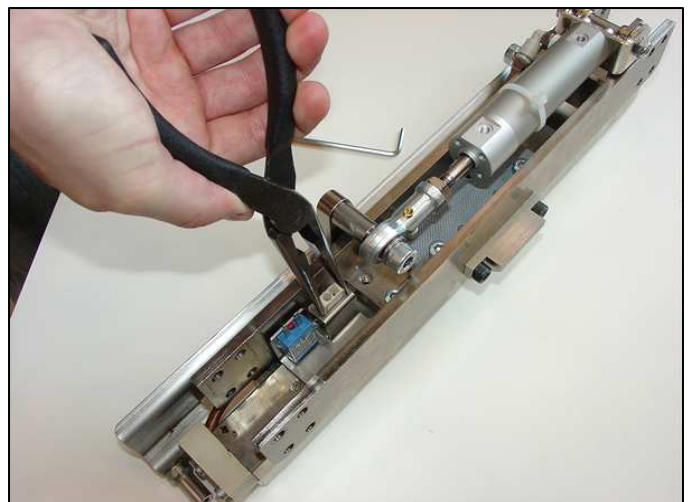


Photo 50: Compressing the white power connector

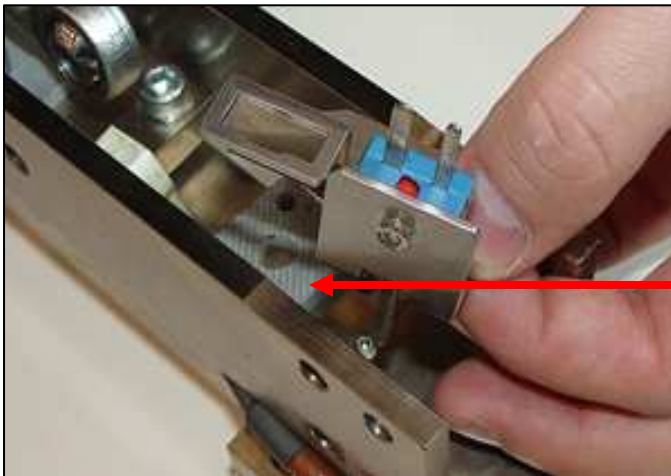


Photo 51: Removing the old sensor and bracket

47. Remove the sensor Bracket

Note: Do not loose this spacer.
It prevents heat transfer to the plastic connector.



Photo 52: Removing the old sensor from the bracket

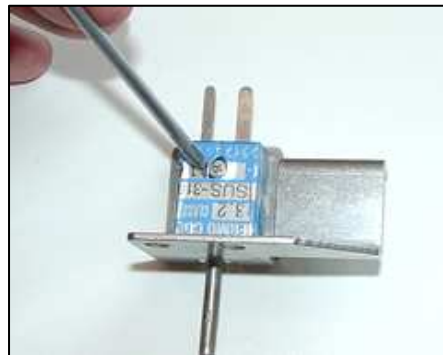


Photo 53: Removing the old sensor from the bracket

48. Remove the old Sensor

Using an a small adjustable wrench and Phillips screwdriver to remove the sensor.

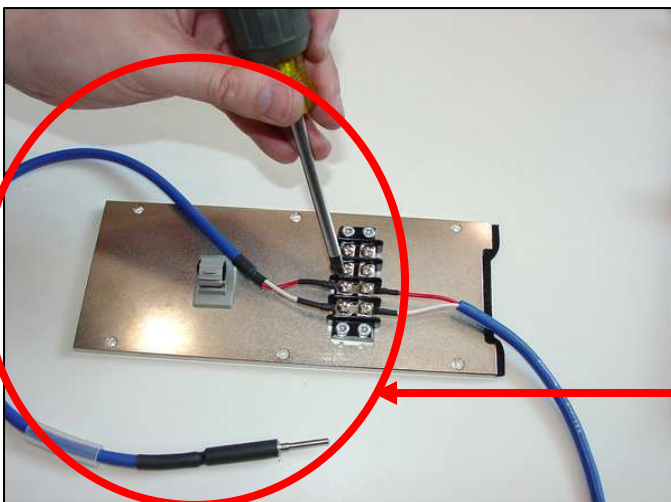


Photo 54: Removing the new sensor

49. Remove the New Sensor from Cover Plate

Using a Phillips Screwdriver to loosen the screws on the terminal block and then remove the New Sensor.

50. **Place the White Connector through the bracket**

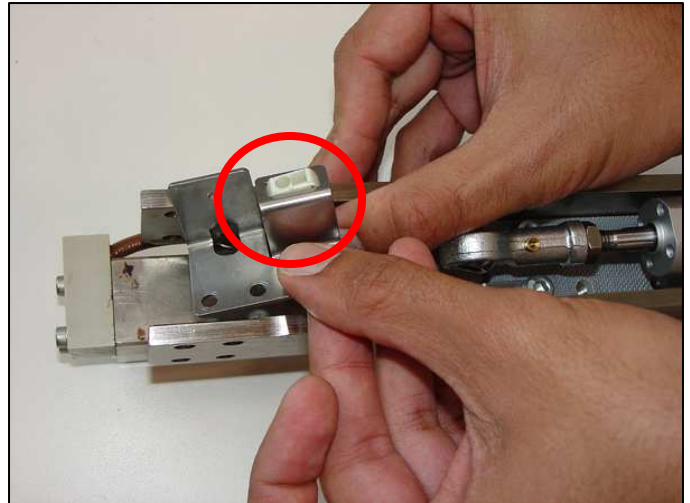


Photo 55: Re-install the white power connector

51. **Slide the Spacer into Position**

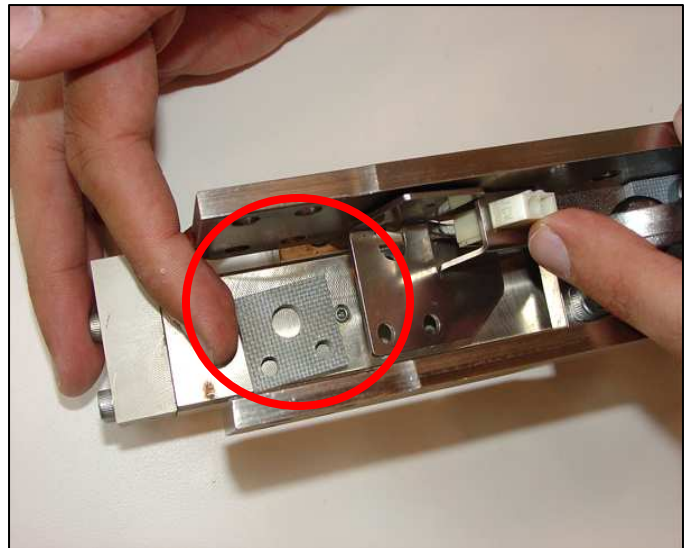


Photo 56: Inserting the spacer

52. **Attach the Bracket to the Jaw Unit**

Using a 3mm Allen wrench,

install the original bolts.
Note: Keep the bolts loose in order to install the new sensor in the next step.

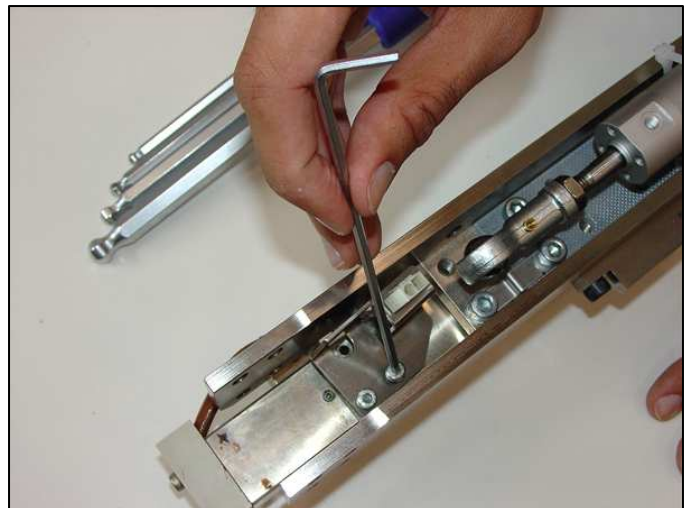


Photo 57: Securing the bracket in place

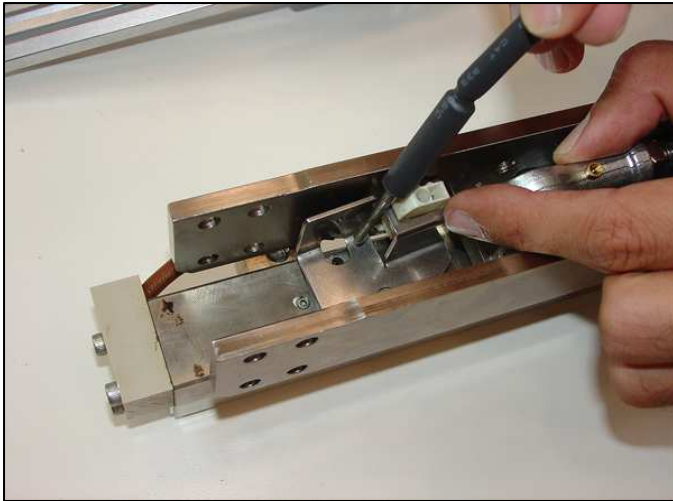


Photo 58: Installing the new sensor

- 53. Install the New Sensor into the Jaw**
Insert the metal tip through the bracket into the jaw.



Photo 59: Checking the sensor position

- 54. Verify that the metal tip is into the Jaw**

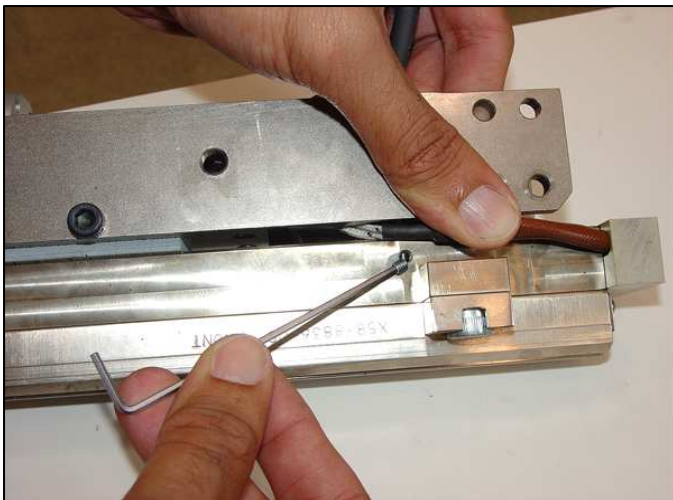


Photo 60: Securing the sensor

- 55. Install the set Screw firmly**
Use a 2mm Allen wrench and do not over-tighten.

56. Tighten the Bracket Bolts:

Use a 3mm Allen wrench.

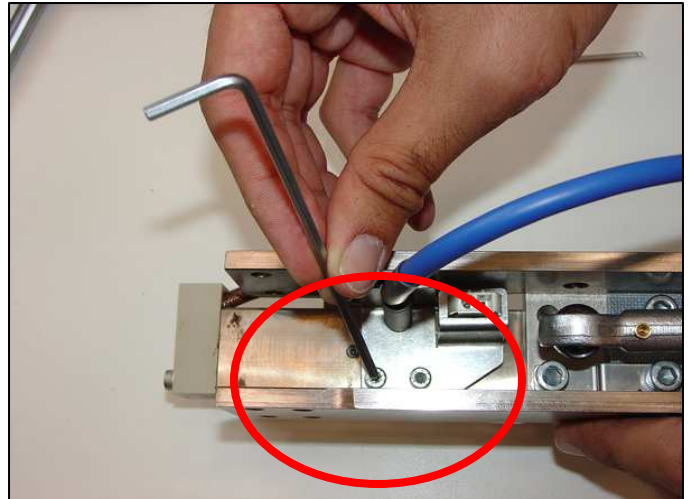


Photo 61: Tightening the bracket bolts

57. Re-install the Knife Bracket

Use a 6mm Allen wrench.



Photo 62: Installing the knife bracket

58. Re-install the Cylinder Bolt

Use a 6mm Allen wrench.

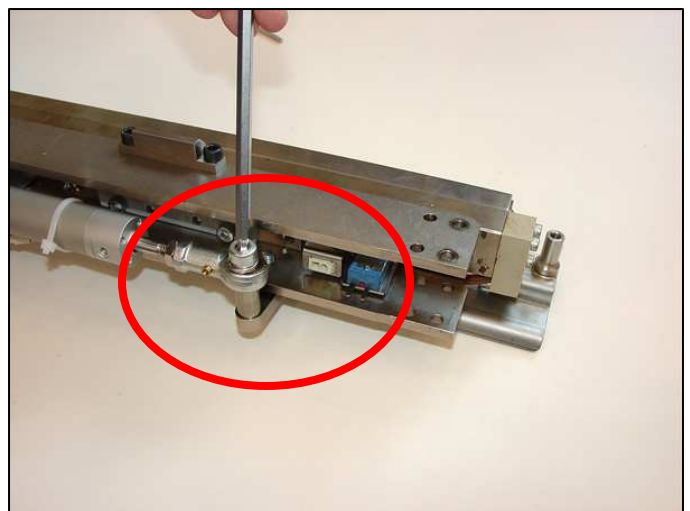


Photo 63: Installing the knife cylinder

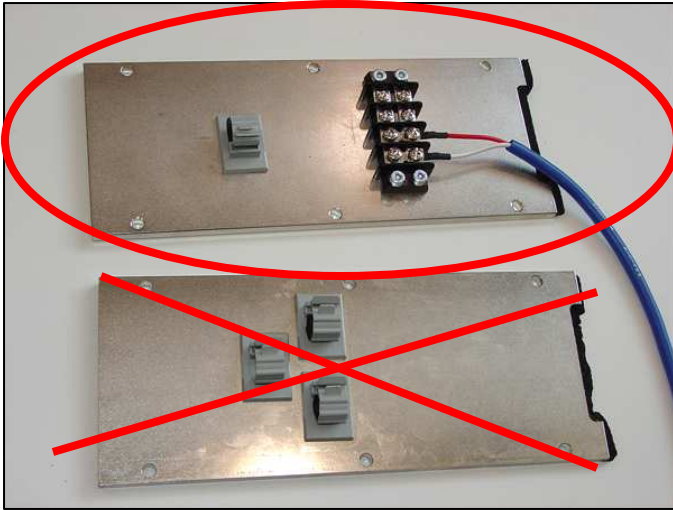


Photo 64: Using the new cover plate

59. Use the New Heater Cover Plate

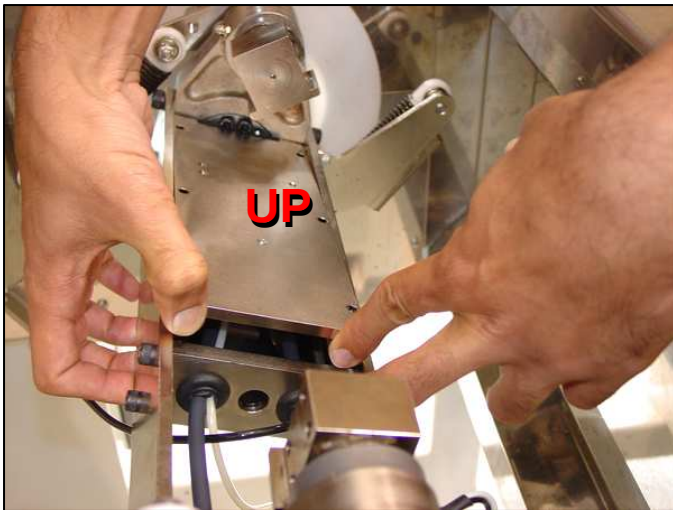


Photo 65: Installing the new cover plate

60. Firmly Position the Heater Cover Plate

Note:

After installing the Heater Cover Plate on the front Jaw. Rotate the Jaw Unit 180 degrees and install the Heater Cover Plate on the Rear Jaw.

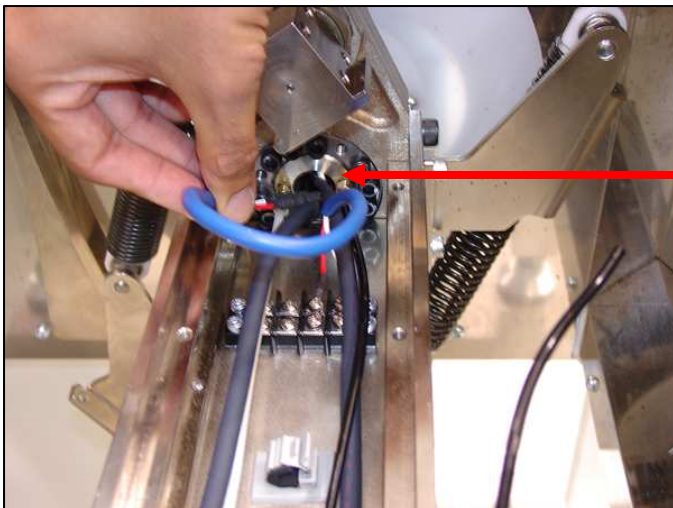


Photo 66: Feeding the sensor and power cables

61. Feed the Front Sensor Cable and Power Cable into the Commutator Housing.

Note: After rotating the jaws, the cables will be upward.

62. Rotate the Jaws and Feed the Rear Sensor Cable into the Commutator Housing.

63. Re-install the Front Jaw.

Note: This jaw unit must contain the knife bracket and the new sensor. and make sure that the cover plate is up.

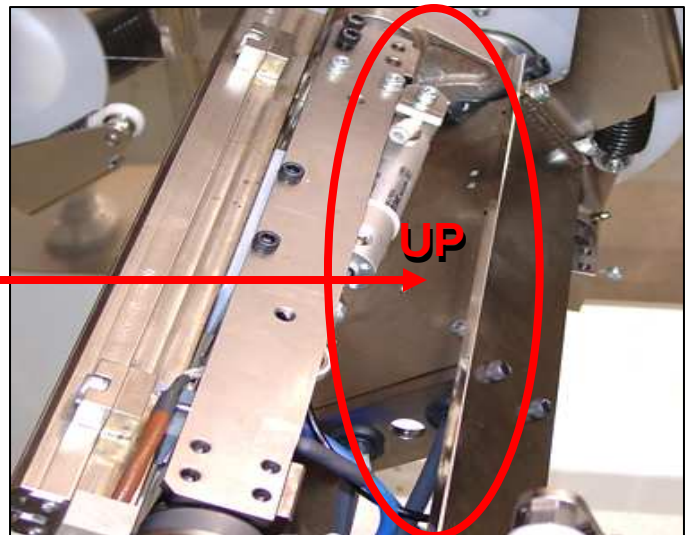


Photo 67: Installing the front jaw

64. Position the Heat Insulator 3 inches from the black end piece.

Photo 68: Positioning the hear insulator

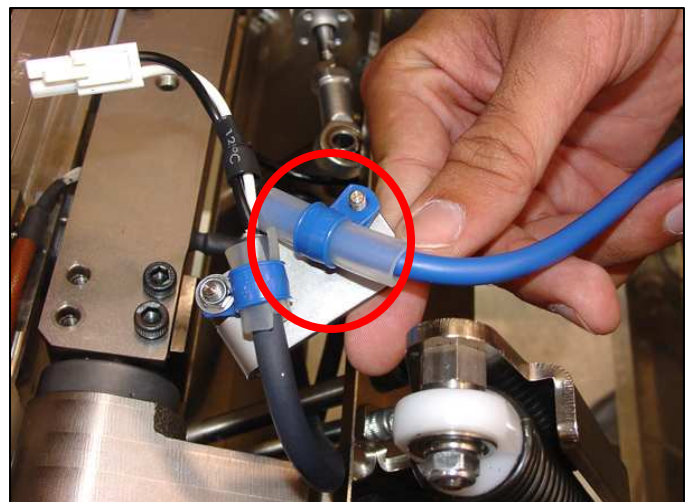
65. Attach the Sensor Cable to the Bracket.
Be sure that the blue clip is in the center of the heat insulator.

Photo 69: Attaching the new sensor cable

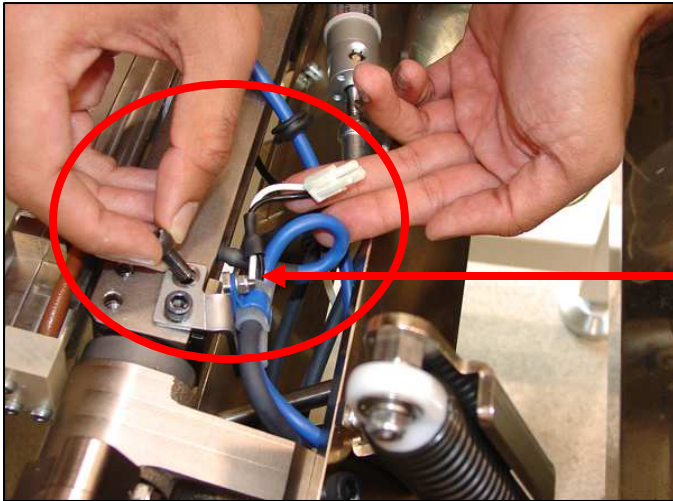


Photo 70: Attaching the bracket to heater

66. Loop the Sensor Cable and Attach the Bracket.

67. Connect the White Power Connector..
Use a 5mm Allen wrench.



Photo 71: Routing cable

68. Route the Cable Under the Round Bar.

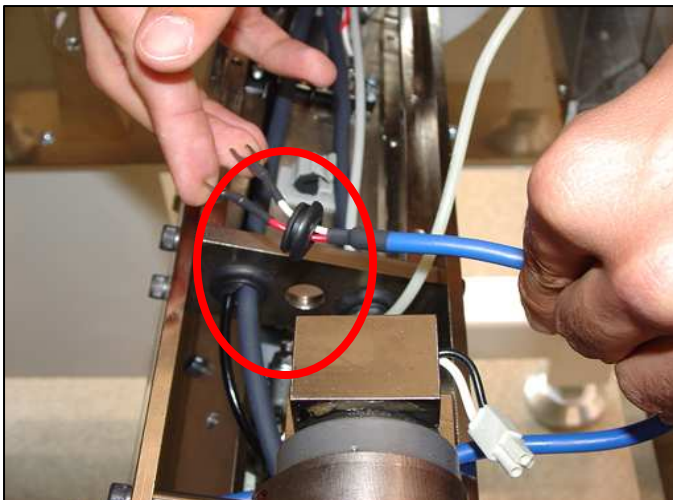


Photo 72: Routing cable through metal bracket

69. Insert the Cable Through the Black Rubber Grommet and Through the Center Hole of the Metal Bracket.

70. Connect the Sensor Cable to the Terminal Strip.

Use a Phillip screwdriver.

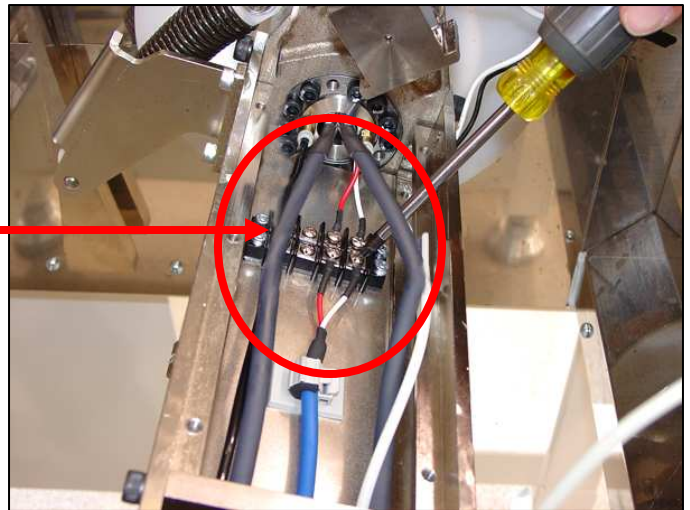


Photo 73: Connecting the new sensor

71. Move to the Rear Jaw

Rotate the Jaws and repeat steps Number 63 through 70.



Photo 74: Connecting the new sensor on rear jaw

72. Connect All Cables to the Front and Rear Commutators and re-install all remaining components.



Photo 75: Connecting commutator wire

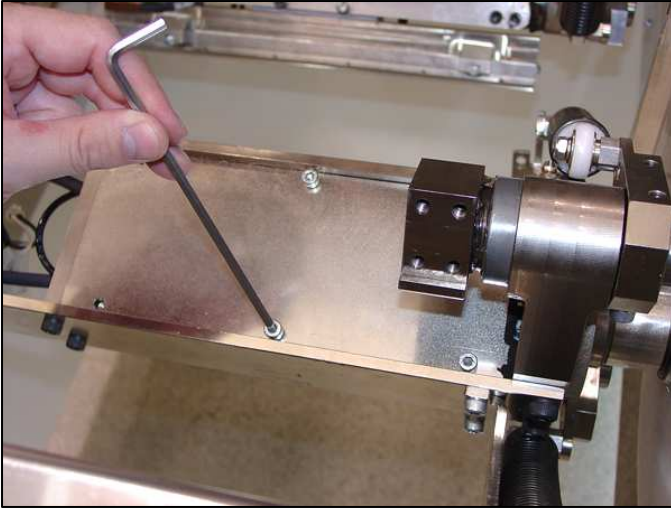


Photo 76: Installing cables cover plates

73. Replace the Cables Cover Plate

Note: Use caution to avoid pinching cables.
Use a 4mm Allen wrench.



Photo 77: Checking jaw alignment

74. Install the last two Jaws.

75. Install the two Knives.

76. Jaw Alignment Required



Photo 78: Installing side panels

77. Install the Front and Rear Stripper Plates.

78. Install the Front and Rear Crumb Plates.

79. Reconnect the Jaw Motor.

80. Install Side Panels.